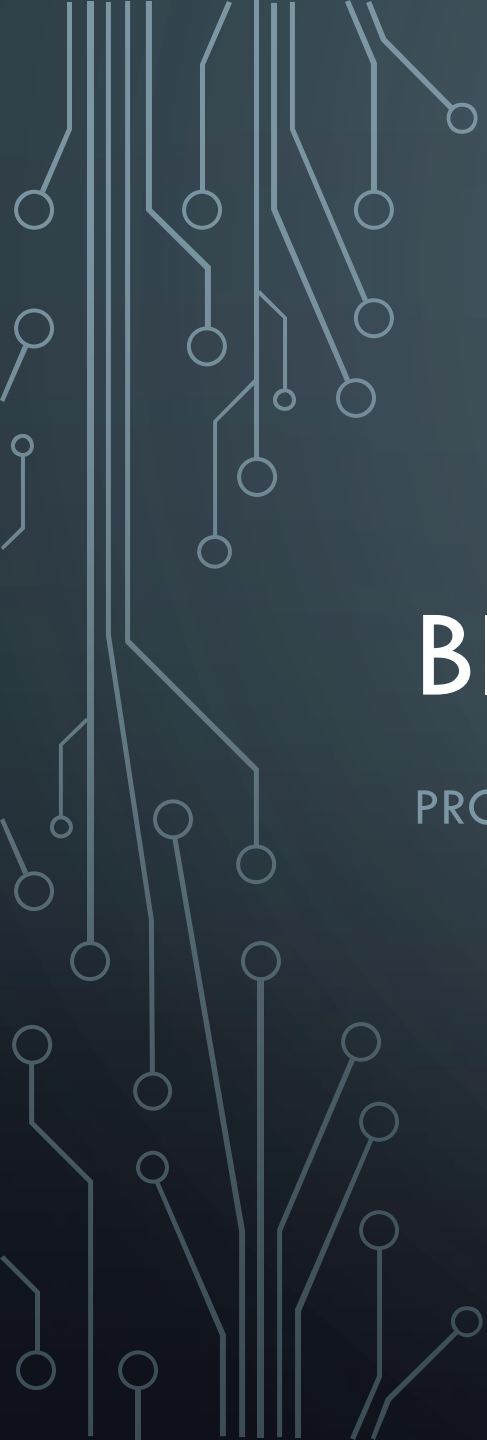


An abstract graphic on the left side of the slide, consisting of a network of white lines and circles on a dark blue background. The lines are vertical and horizontal, with some branching out, resembling a circuit board or a data network. The circles are of varying sizes and are placed at the ends of the lines, creating a sense of connectivity and flow.

BIG DATA

PROCESSING

LEARNING OBJECTIVES

- At the end of this presentation, you should be able to
 - Explain data processing in different scenarios
 - Explain ETL, its steps and purpose
 - Explain the procedures and purposes of batch and stream processing
 - Discuss data processing methods for structured and unstructured data
 - Name a few Hadoop tools for different data processing steps

WHAT IS BIG DATA PROCESSING?

- the collection and manipulation of data to produce meaningful information



ETL



data
lakes



DATA PROCESSING

BATCHES

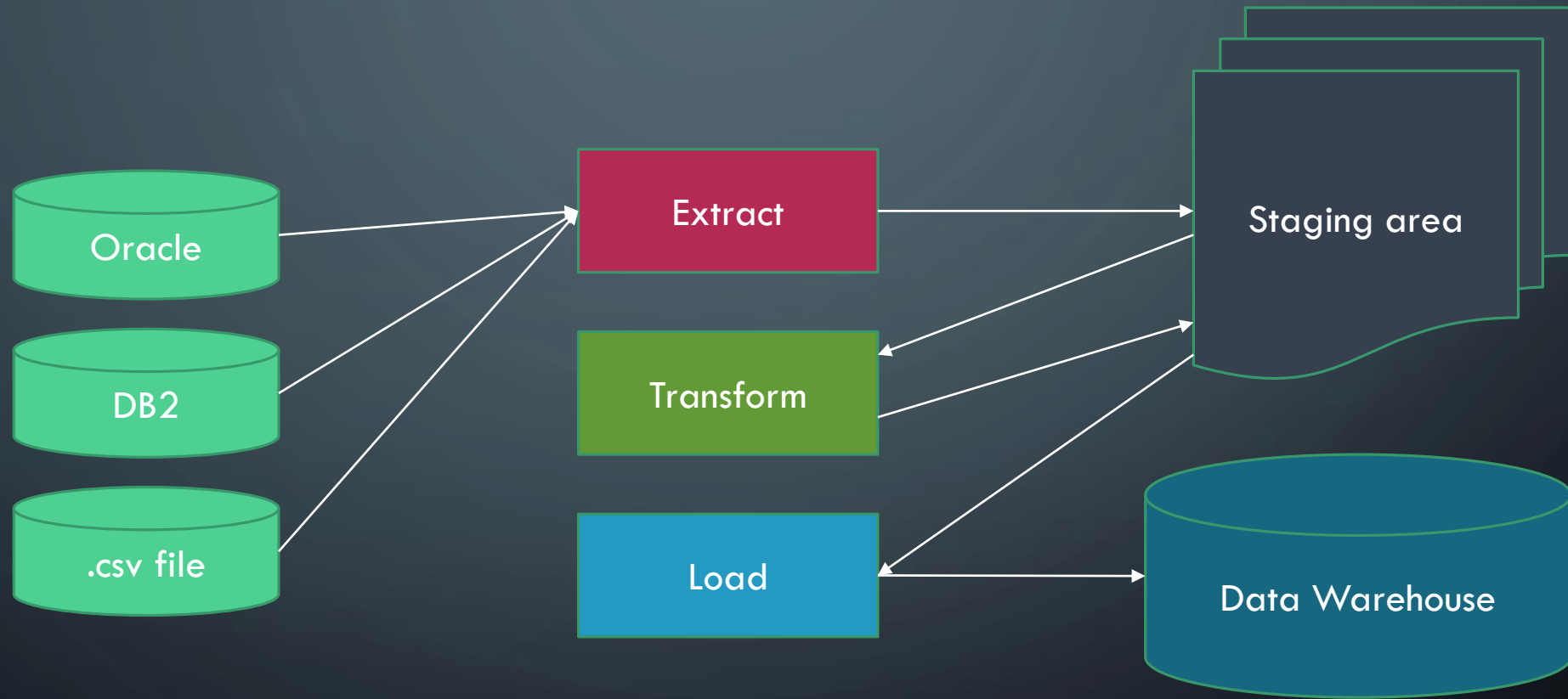
- Periodic
 - Irregular and unpredictable arrivals
- Scheduled
 - Predictable and timed arrivals



STREAMS

- Real-time processing
 - Data has to be acted on in a matter of milliseconds
- Near real-time processing
 - Data has to be dealt with in a matter of minutes

ETL

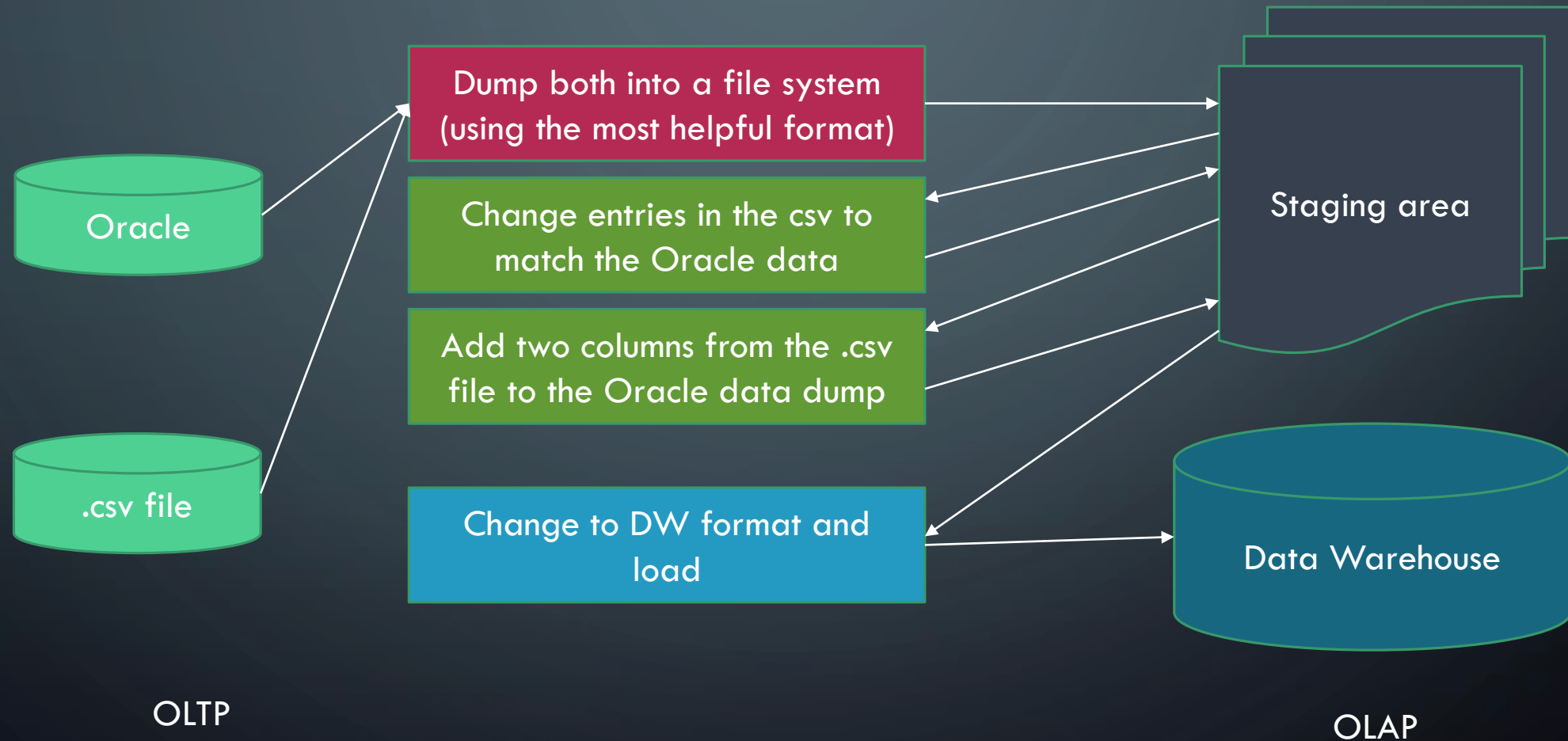


OLTP

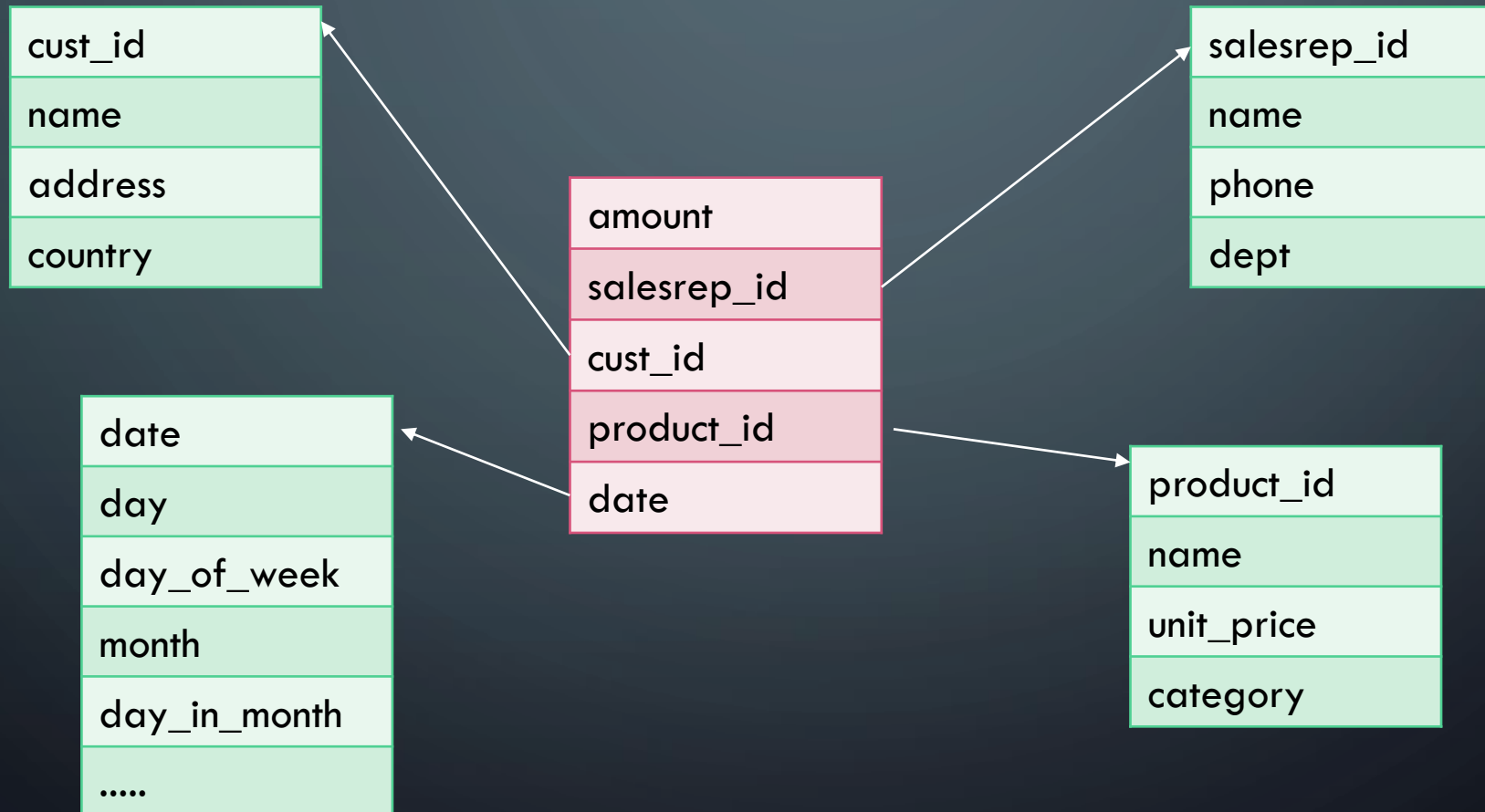
OLAP

5

ETL - EXAMPLE



TRADITIONAL DATA WAREHOUSE



DATA INTEGRITY

Cleaning and structure



PROBLEMS WITH DIRTY DATA

- May affect the interaction with third parties
 - Spelling a customer's address wrong
- Wrong results from queries and analyses
 - Data may be missed
 - Data may be misinterpreted
- Results from queries and analyses may produce inconsistencies
 - Databases optimised for speed may produce conflicting information



DATA INTEGRITY

- Integrity is trustworthiness, veracity
- Have to define – what is clean data?
- What are the potential errors in new raw data?
 - How can the issues be prevented/eliminated during ETL?
- When do we stop needing the data?
 - Have to archive data that is no longer needed.



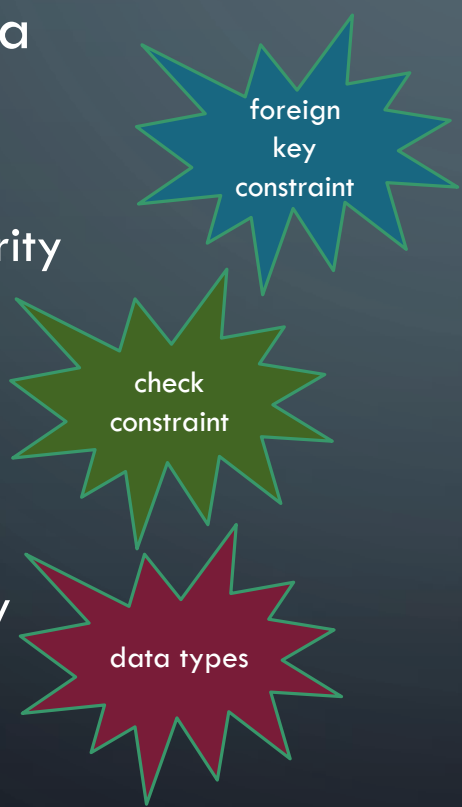
INTEGRITY IN RELATIONAL DATABASES

- Database Schema

- Relational integrity

- Entity integrity

- Domain integrity



foreign
key
constraint

check
constraint

data types

- Isolation levels

- Transaction design (OLTP!)

- Synchronisation (distributed!)

- 'Two-phase commit'

ACID

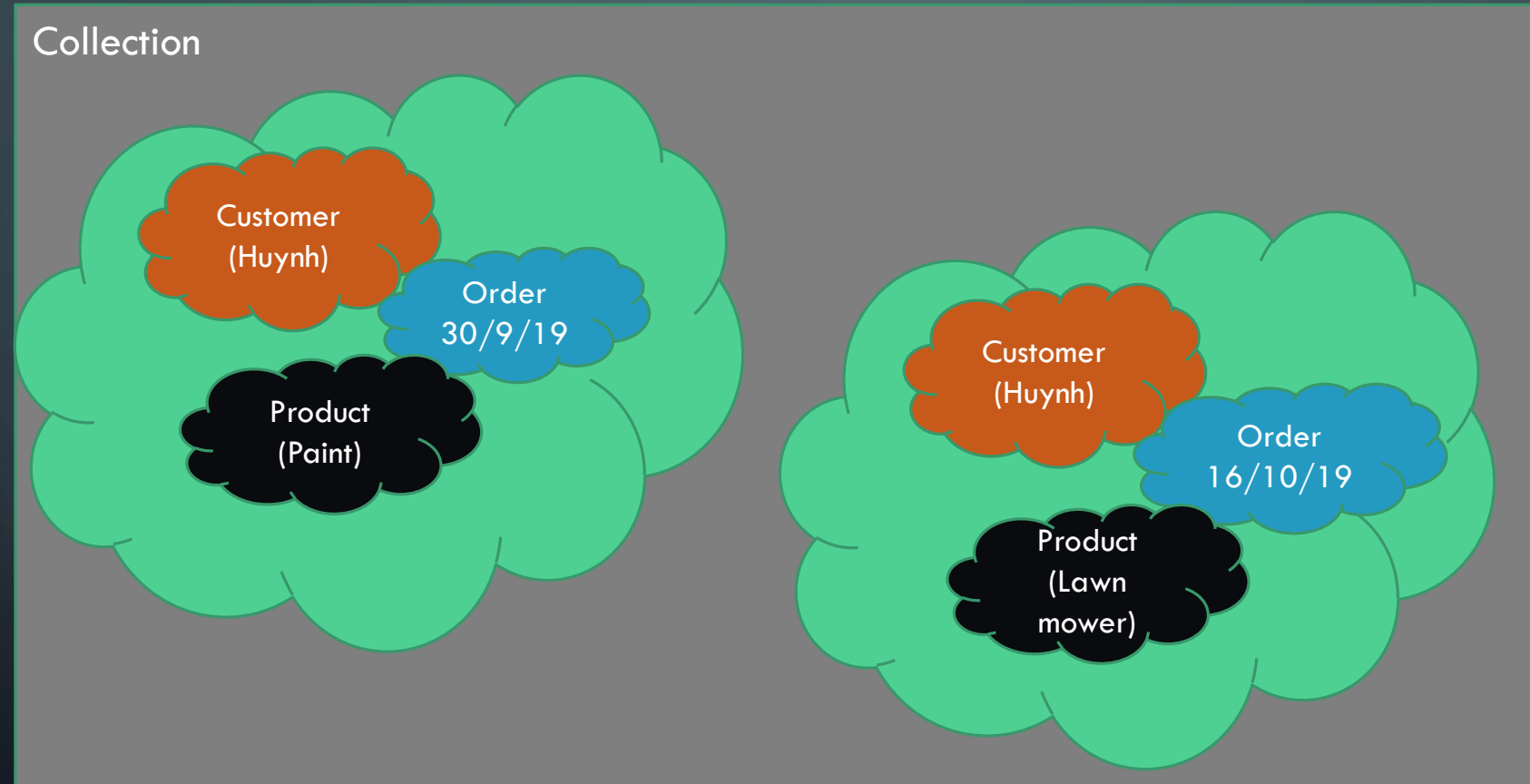
INTEGRITY IN STRUCTURED AND SEMISTRUCTURED DATABASES

- Basically Available
- Soft State
- Eventually Consistent

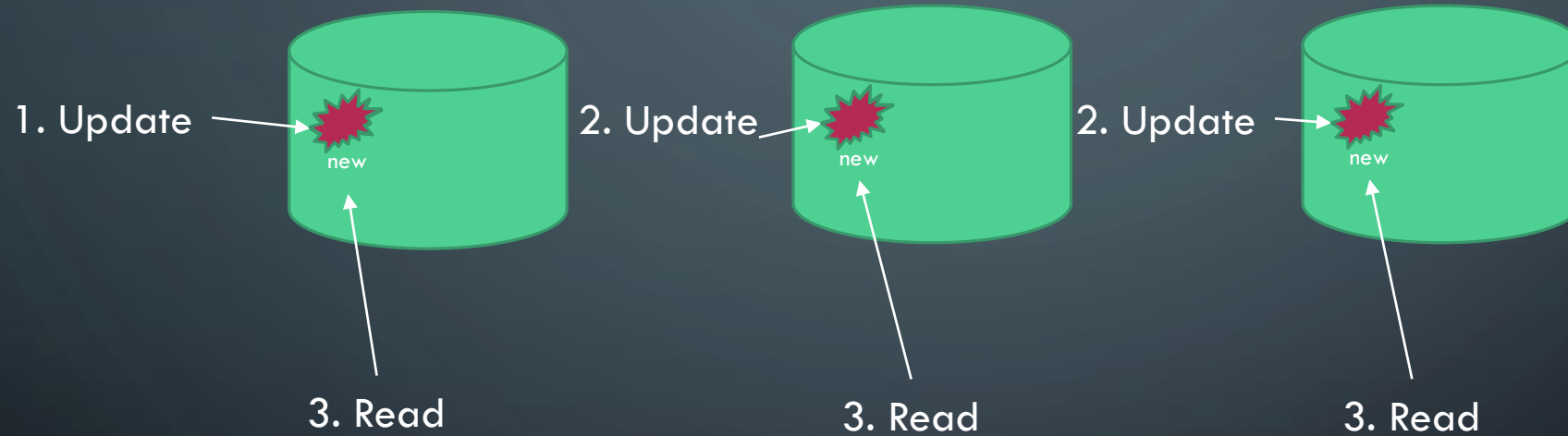
ACID Compliant	BASE Compliant
Strong consistency	Weak consistency
Strong isolation	High availability
Committed results	Immediate results
Conservative availability	Aggressive availability

BASE

SCHEMA IN NON-RELATIONAL DATABASES

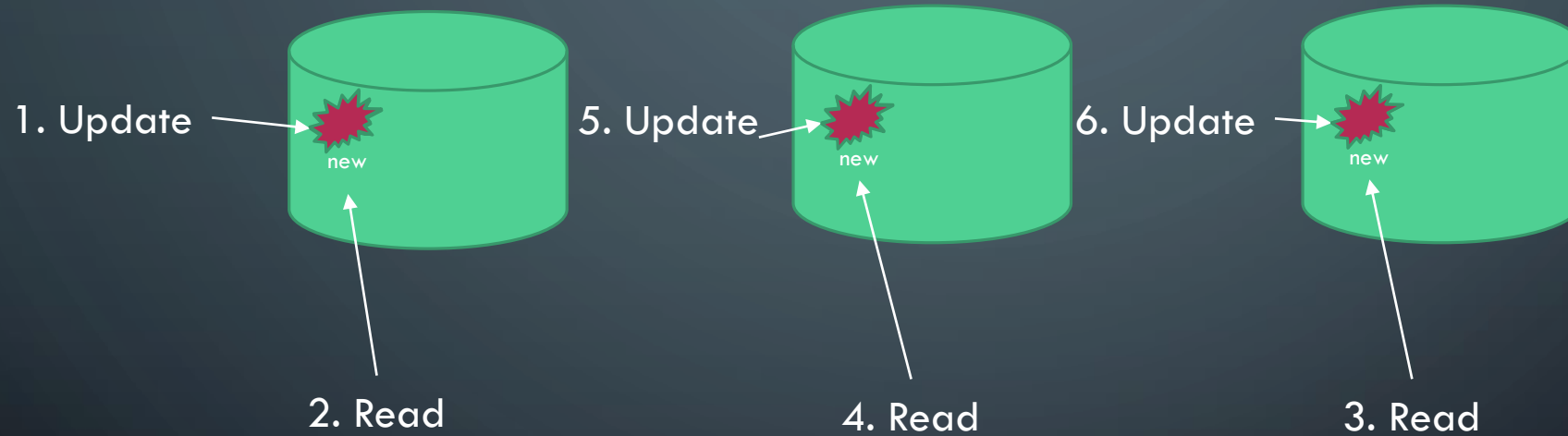


ACID-COMPLIANT BEHAVIOUR



Several copies of the same database.

BASE-COMPLIANT BEHAVIOUR

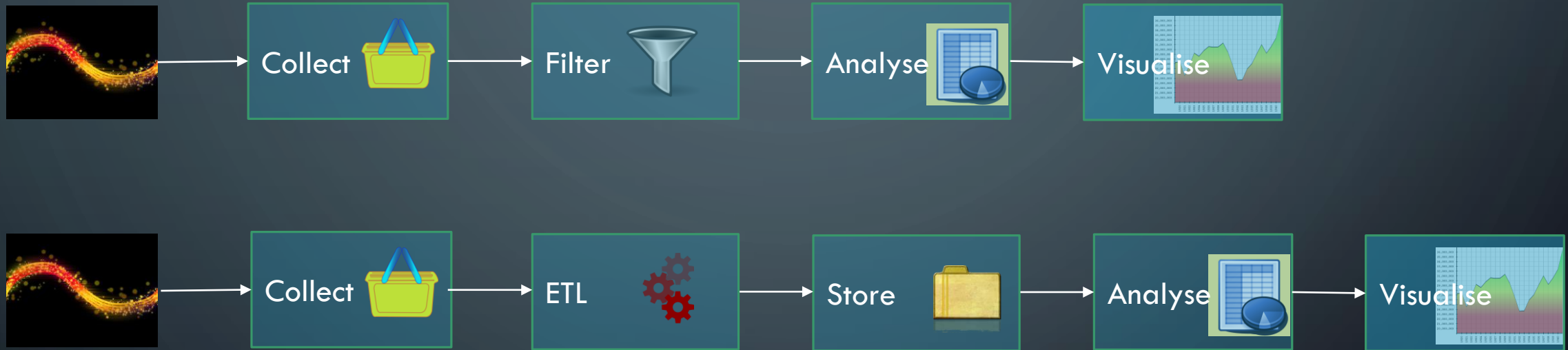


Several copies of the same database.

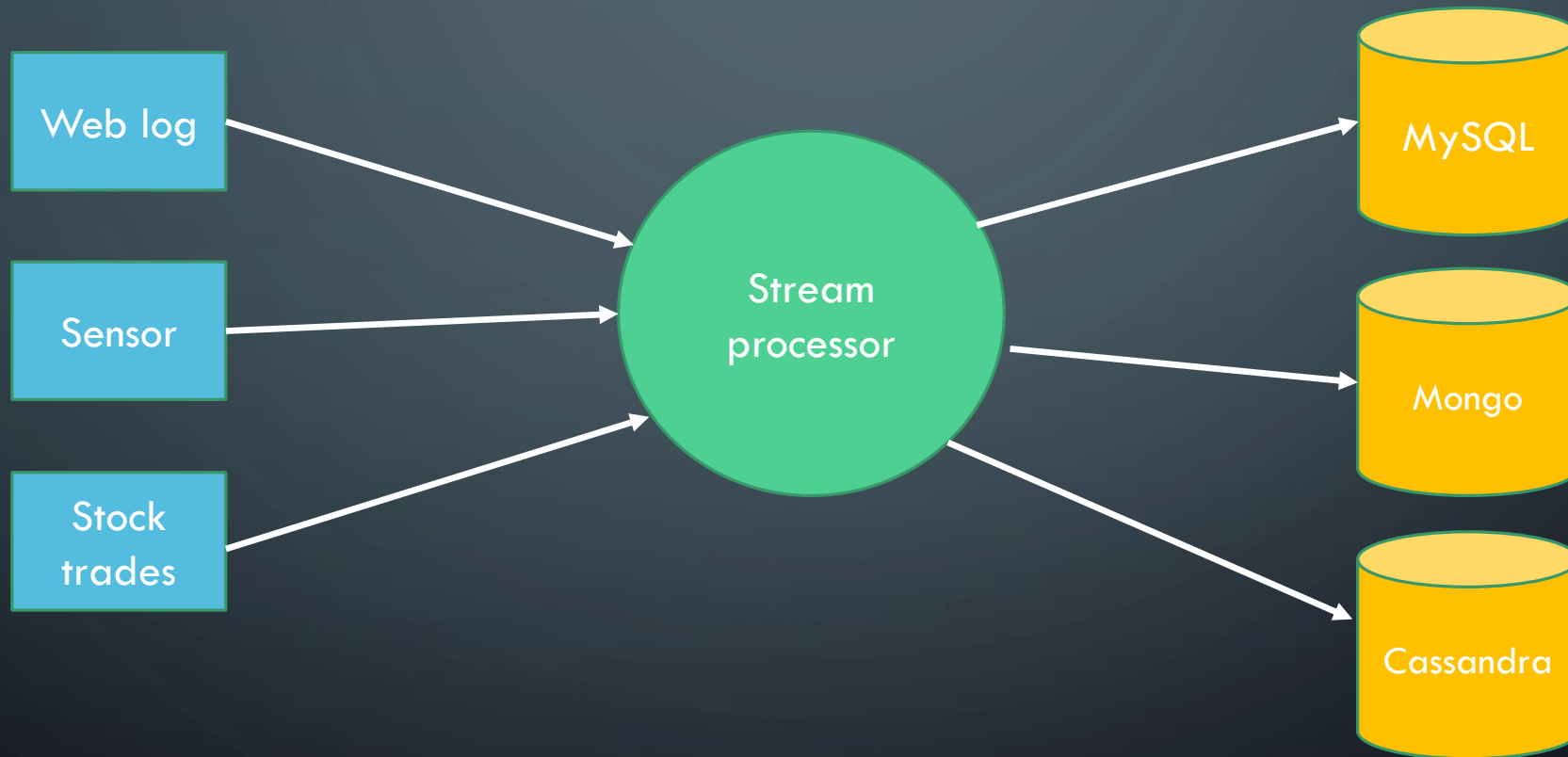
STREAM PROCESSING



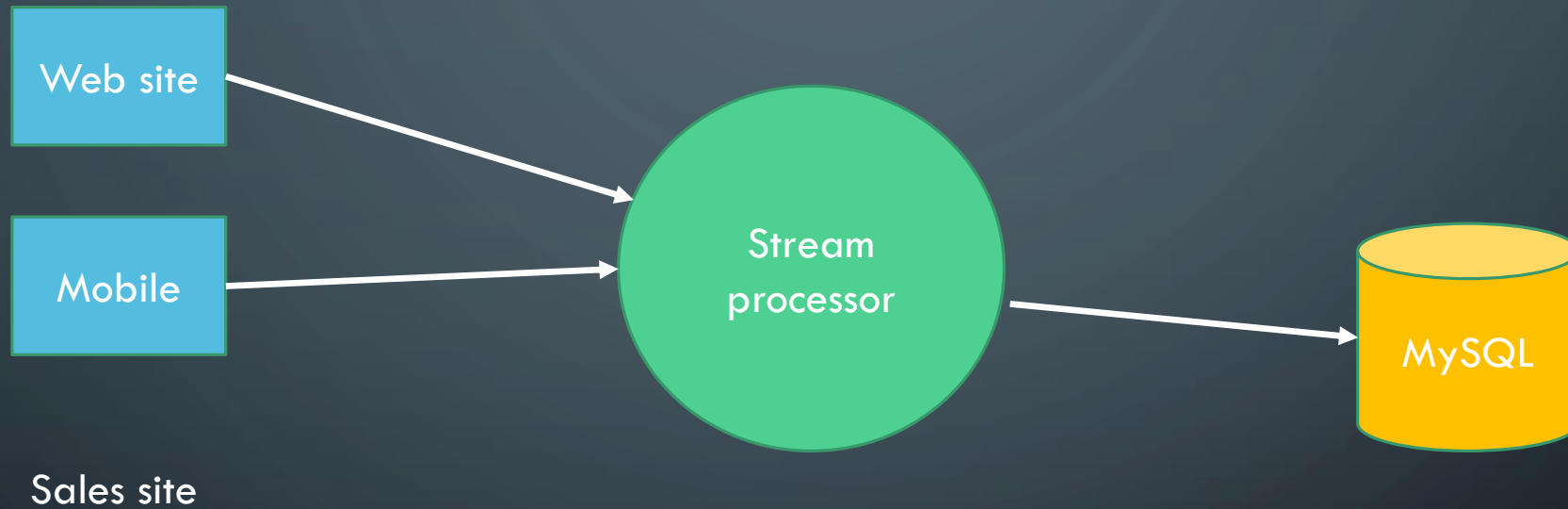
STREAM PROCESSING



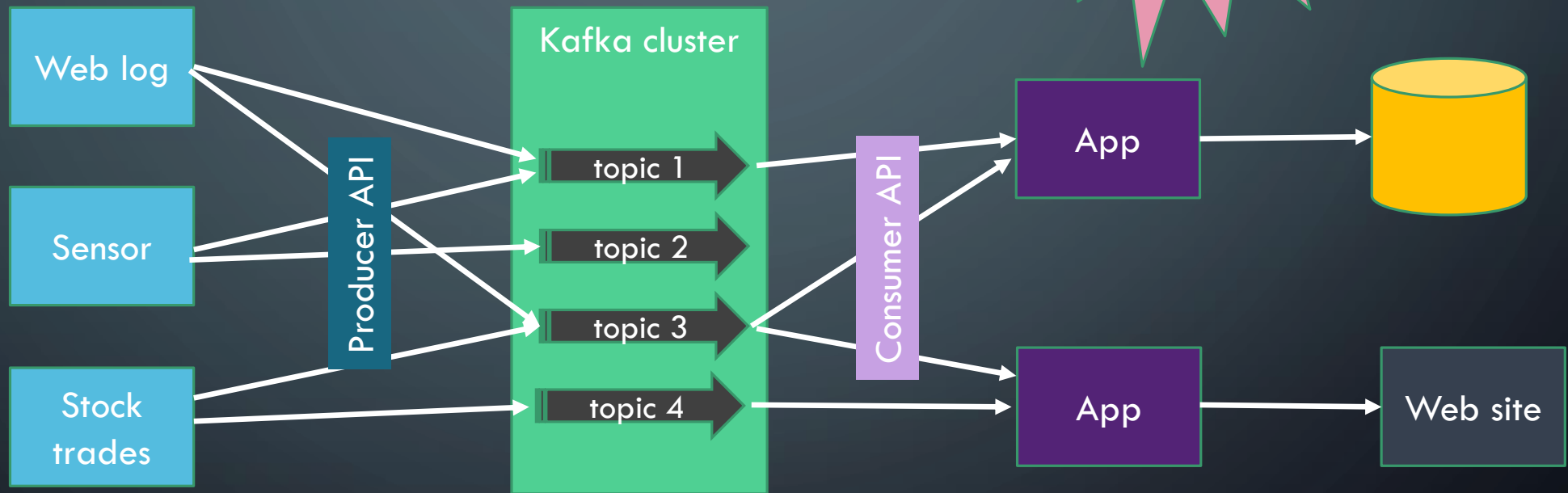
TYPICAL STREAM PROCESSING SCENARIO



LESS TYPICAL STREAM PROCESSING SCENARIO



APACHE KAFKA

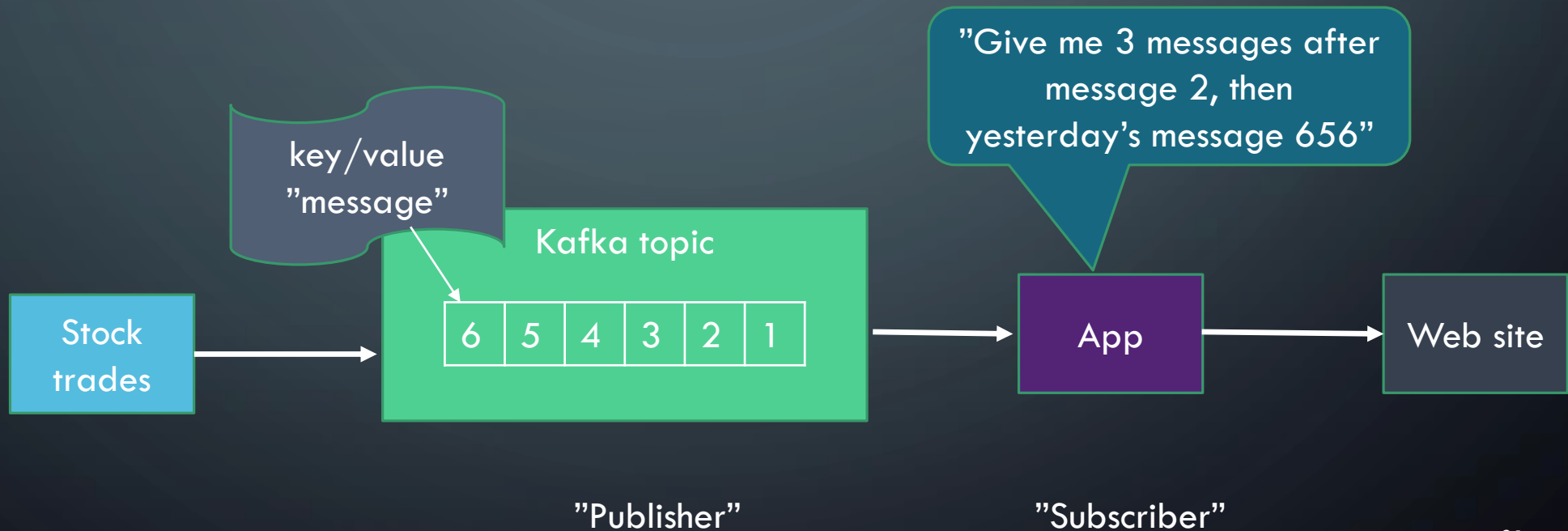


Producer

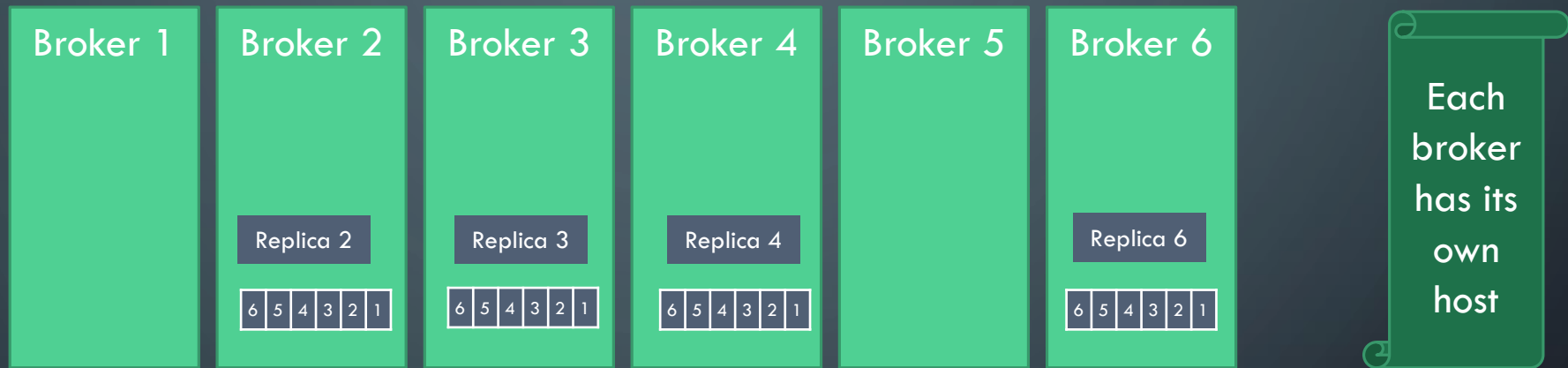
Consumer

20

APACHE KAFKA TOPIC



KAFKA TOPICS AND PARTITIONS

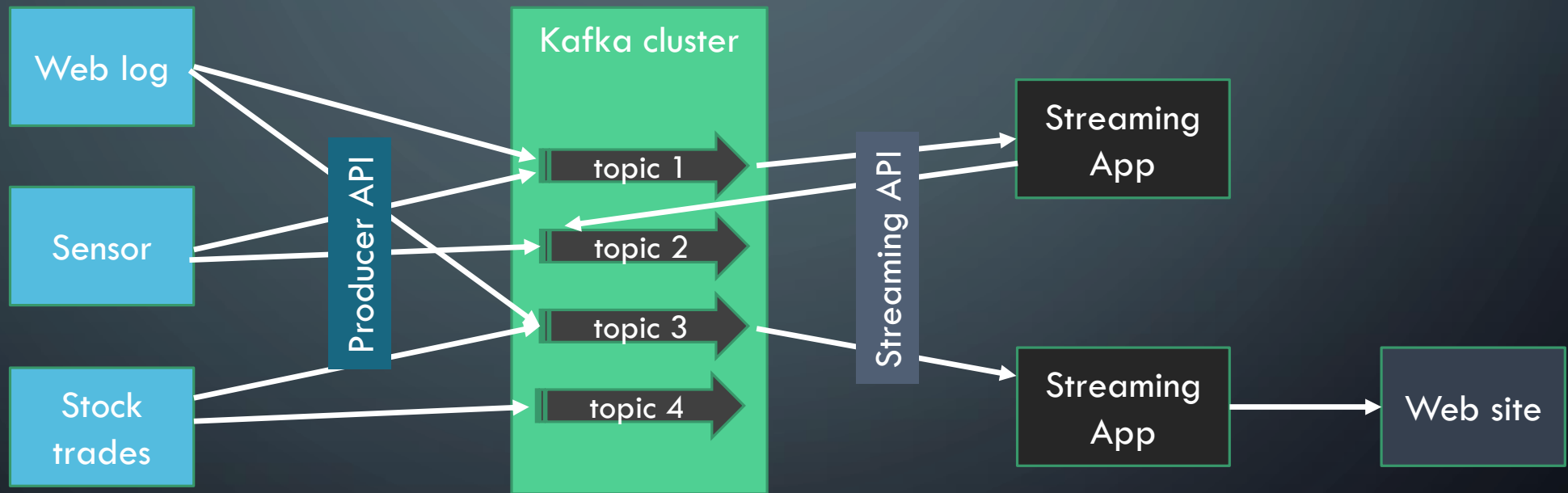


Topic:
3 partitions



4 replicas of topic 1
numbered after brokers

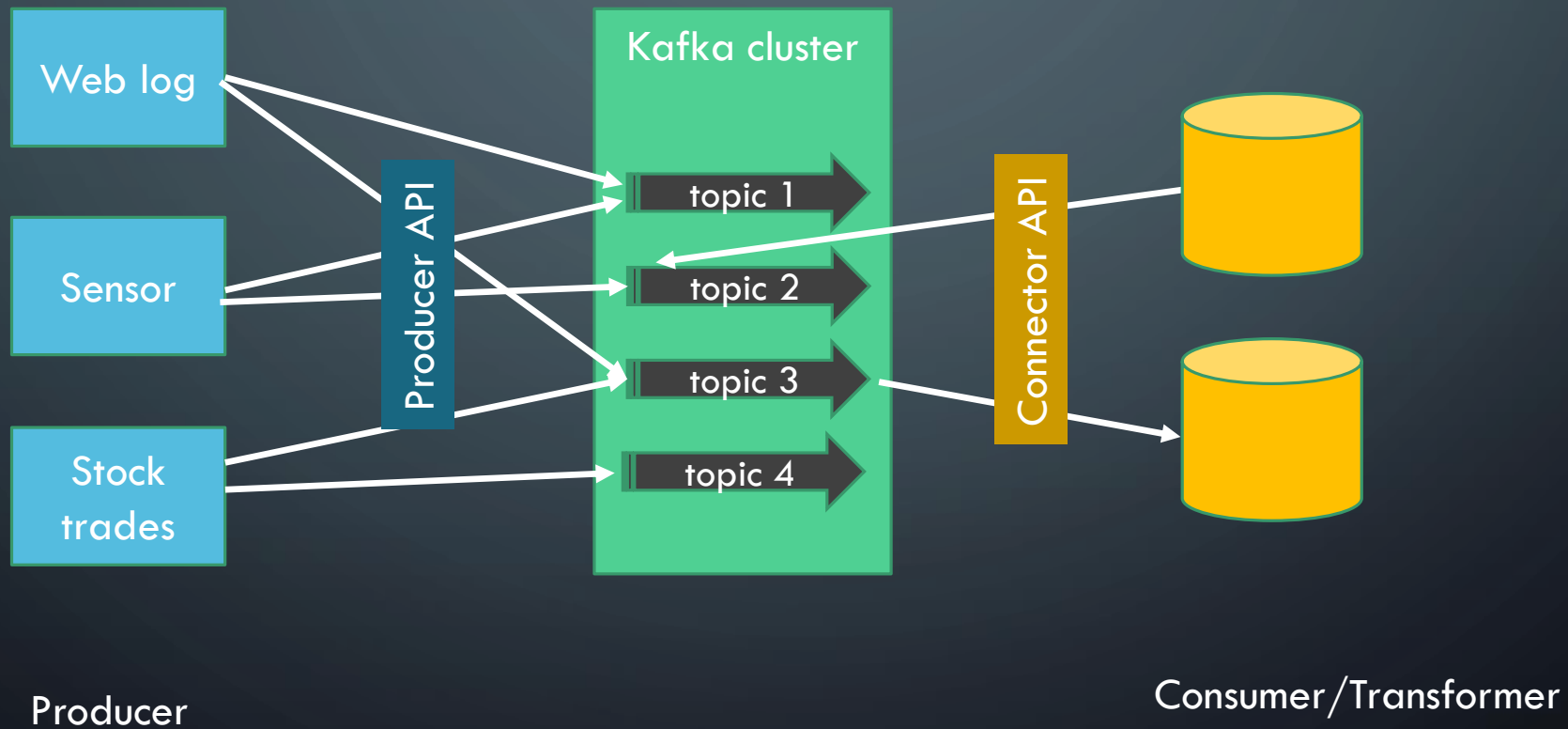
APACHE KAFKA



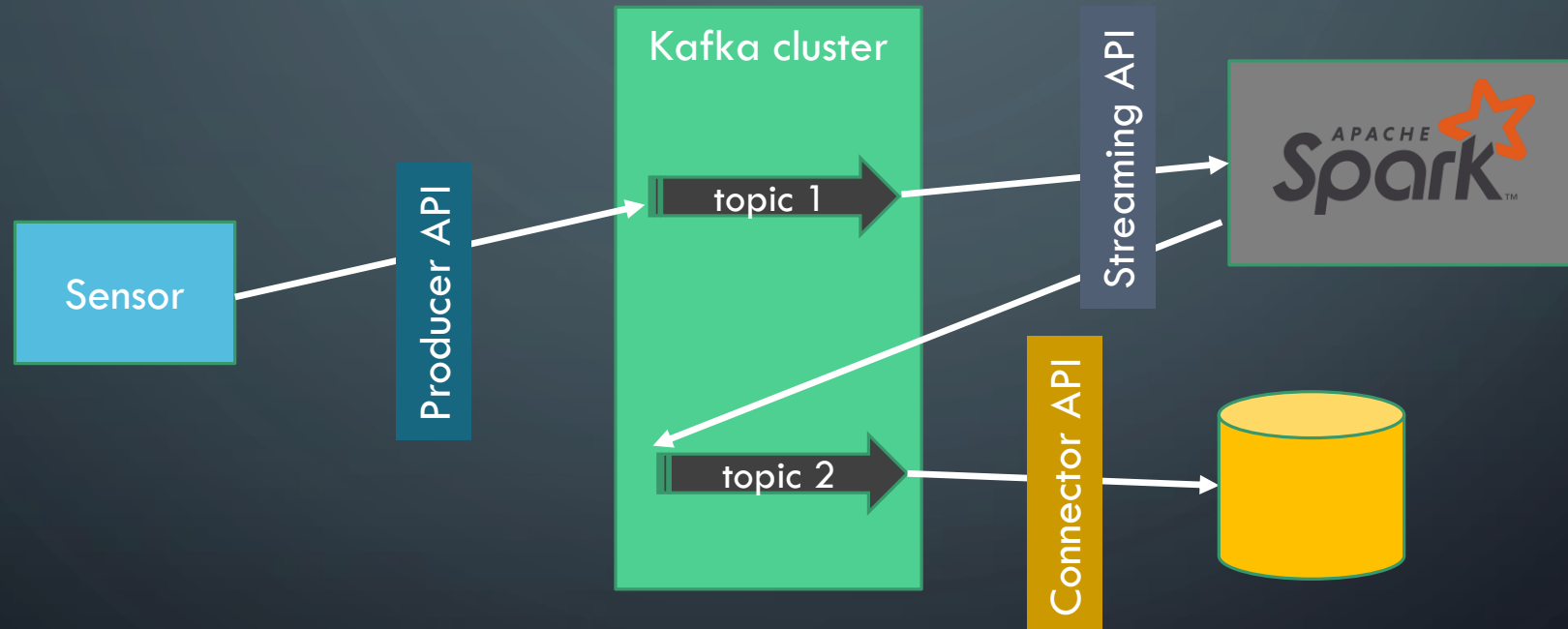
Producer

Consumer/Transformer

APACHE KAFKA



APACHE KAFKA AND SPARK



APACHE SPARK

Spark Core
and
Resilient
Distributed
Datasets

Spark SQL

Spark
Streaming

Machine
Learning
Library
(MLlib)

GraphX



BATCH PROCESSING



APACHE SPARK

Spark Core
and
Resilient
Distributed
Datasets

Spark SQL

Spark
Streaming

Machine
Learning
Library
(MLlib)

GraphX



AND ALSO: PIG AND HIVE!

PIG



HIVE

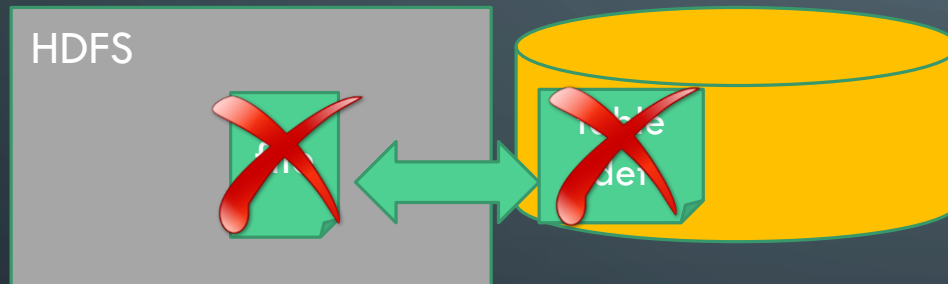


- Can take a delimited (csv) file and treat it like a table.
- Pig also has efficient ways of dealing with unstructured data

HIVE: MANAGED AND EXTERNAL TABLES

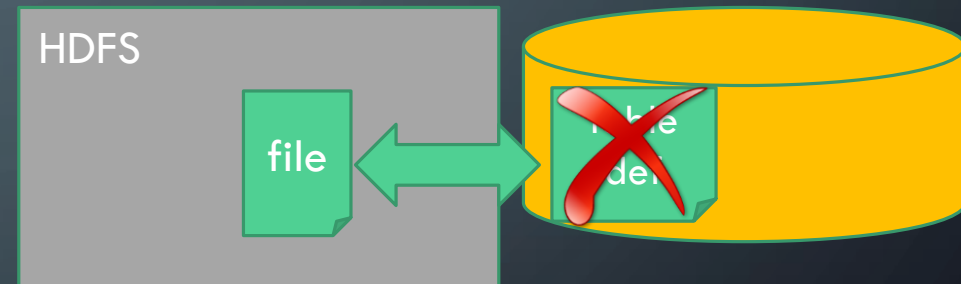


MANAGED



```
CREATE TABLE <tablename>  
(col1 datatype, col2 datatype)  
ROW FORMAT DELIMITED FIELDS TERMINATED BY '\t'  
STORED as textfile;
```

EXTERNAL



```
CREATE EXTERNAL TABLE <tablename>  
(col1 datatype, col2 datatype)  
ROW FORMAT DELIMITED FIELDS TERMINATED BY '\t'  
STORED as textfile;
```

INGESTING DATA



SUMMARY

- Data processing means collecting and manipulating data so that it becomes meaningful.
- ETL describes data processing from raw to storage, including cleaning and structural changes.
- Data can be processed in batches or streams (or both).
- Data integrity is often described in terms of ACID-compliant behaviour, whereas BASE-compliant behaviour describes a high-throughput approach associated with NoSQL databases.
- In-memory processing is an alternative to databases.
- There are many convenient ETL tools in Hadoop for ETL.